

An authentication scheme for FANET packet payload using data hiding

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ABSTRACT

Flying ad hoc networks (FANETs) are gaining much traction in recent years due to their wide range of applications, resulting in the emergence of a number of studies on the facilitation of their widespread adoption. One of the key areas being investigated is the establishment of security strategies to combat the large amount and variety of security threats that they inevitably face when they are operating in the field. Since a majority of FANET's operations involve the transmission of network packets containing both instructions and sensitive data, the protection of these packets is critical. This paper proposes an authentication scheme to protect the payload of the packets being transmitted within the FANET. Since FANETs are highly resource-constrained, we opted for a lightweight and energy-efficient approach. The proposed scheme essentially encodes the payload of a physical layer data frame into a compact bit stream, which is subsequently embedded into the cyclic prefix (CP) at the physical layer. The encoding process includes the XOR-operation and JBIG2 lossless compression. At the receiver's end, the hidden authentication data is extracted and decoded for comparison against the payload of the received data frame. Two enhancements are introduced to further enhance robustness against masquerade attack. The proposed scheme is evaluated in terms of bit error rate (BER) under different signal-to-noise ratio (SNR) settings. In all considered SNR settings, the proposed authentication scheme achieves BERs of $< 0.7 \times 10^{-4}$ when 7-bit Hamming code is implemented. Experiment results also confirm that proposed scheme is able to localize the tampered data frames.

1. Introduction

In recent years, the use of Unmanned Aerial Vehicles (UAVs) has become commonplace in both civilian and military domains due to their accessibility, versatility and low operational cost. Applications for these UAVs include parcel delivery [1], crowd and traffic monitoring [2], search-and-rescue missions [3], and disaster management [4]. The decentralized and ad hoc manner in which the UAV nodes communicate ensure that the network remains functional even when a node malfunctions. These features enable UAVs to form their own communication links, which is especially crucial during critical missions. This communication architecture has led to the coining of the term *Flying Ad hoc Network* (FANET), which is commonly used when referring to these networks [5]. FANETs typically operate in the field and are vulnerable to various security threats, including masquerading, eavesdropping, modification of network packets, to name a few [6,7]. Potential adversaries may even modify packet payloads to insert false instructions in the hopes of hijacking the UAV nodes. Therefore, security solutions need to be put in place to combat such attacks and protect the contents of the network packets being circulated in the network.

Presently, many schemes are designed to protect message integrity in network systems, including message authentication code (MAC)

schemes [8,9], digital signatures [10–12] and, recently, machine learning-based schemes [13,14]. In addition to user authentication, researchers have been looking into protecting the payload of individual data frames, be it through payload encapsulation or intrusion detection through the generation of payload signatures [15–17]. However, FANETs are highly constrained in terms of storage, computational power and battery life. Therefore, a lightweight and energy-efficient approach under restricted resources and time constraints is highly desired instead of using complex cryptographic protocols. Likewise, some recent works have proposed the use of cryptographic protocols in a lightweight and efficient manner to better suit FANET requirements.

Rajasekaran et al. [18] suggested an authentication scheme that utilizes a simple cryptographic pairing and hashing operations for privacy preservation during mutual authentication. Alladi et al. [19] presented a two-stage mutual authentication protocol for SDN-based multi UAV networks to be deployed in surveillance areas. Tian et al. [7] proposed an authentication protocol that considers both efficiency and security, where a short online/offline signature layout is used to secure a network of small-scale UAVs. Zhang et al. [20] proposed a compact authentication and key agreement scheme that involves a one-way secure hash function where UAVs and users mutually authenticate

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each other. Similarly, Hussain et al. [21] suggested a three-factor authentication scheme. Although these schemes are energy-efficient and lightweight compared to typical cryptographic schemes, they still deal with certain bit-sensitive operations such as hashing. This might not be ideal in a network environment such as FANET, which is extremely prone to bit errors resulting from fast-paced nodes and frequent topology changes.

Data hiding is a form of communication that has existed for some time, and has served as a lightweight alternative when it comes to communication security [22]. Data hiding in networks usually takes advantage of certain loopholes such as anomalies in communication streams [23], unused fields in packets [24] and flexible network protocols [25] to transmit extra data. Therefore, those less computation-intensive loop holes (viz., hiding venues) are favourable for data hiding in networks with resource-constrained nodes such as FANETs.

This paper explores the condensation of data frame payloads through compression and data hiding to verify the payload of data frames received by the nodes of a FANET. Specifically, we want to be able to verify whether the contents of each received packet have not been modified along the communication path. In addition to being extremely resource-constrained, FANETs are also highly dynamic and unstable due to the fast-moving nature of UAV nodes, which result in frequent topology changes. Therefore, it is also of utmost importance that the security solutions are able to account for the bit errors that are inevitable due to continuously refreshing communication links within this dynamic multi-hop network environment. The main contribution of this work is to design a payload authentication scheme through the combined use of masking (XOR), lossless compression technique and data hiding. The proposed scheme is validated using a simulated FANET model which is implemented in MATLAB 2018b. The rest of the paper is structured as follows: Section 2 presents the basics of FANETs and reviews conventional FANET-based data hiding methods. Section 3 details the proposed method with individual descriptions for each major step in the process. Section 4 discusses potential security enhancements to be used in conjunction with the proposed scheme in order to prevent common security attacks. Section 5 presents the experiment results attained by the proposed method and finally, Section 6 concludes this paper and explores potential future directions.

2. Preliminaries and related works

2.1. FANET architecture

A FANET typically consists of a ground control station and at least three to four UAV nodes. Each UAV acts as its own communicating entity and transmits information through the network using wireless multi hop communication based on network connectivity. Therefore, FANET does not need to rely on a preexisting, central infrastructure or a pre-set routing algorithm to function.

FANET follows the IEEE 802.11p standard, which was initially proposed for vehicular ad hoc networks (VANETs) [26]. Similar to many other standards, IEEE 802.11p follows the network open systems interconnection (OSI) model, which consists of seven layers. These layers are: application, presentation, session, transport, network, data-link and physical layer. Specifically, the physical layer of the FANET facilitates signal transmission by performing modulation and signal encoding steps. The data comprising network packets are modulated into sinusoidal waveforms and transmitted through radio propagation using a standard named orthogonal frequency-division multiplexing (OFDM). OFDM is a process that allows the nodes to transmit independent data streams simultaneously through a radio channel by forming OFDM symbols. Interested readers may refer to [27] for a comprehensive treatment of the IEEE 802.11p standard.

Typically, in wireless networks, multi-path propagation is an ordinary phenomenon and commonly results in delayed copies of the transmitted signal due to the reflection from obstacles. This points to

inter-symbol interference (ISI) between the transmitted OFDM symbols. In order to avoid ISI, an interval is introduced in front of each OFDM symbol, which is known as the Guard Interval (GI). According to the IEEE 802.11 standards [28], this interval is filled with a set of bits known as a Cyclic Prefix (CP). The CP usually represents the ending portion of the OFDM symbol that it precedes. However, the CP itself serves no purpose for the transmitted data and is disregarded by network devices [29,30]. CP will be explored further in this work to embed the authentication data.

2.2. FANET-based data hiding methods

Since a large portion of FANET's operation relies on gathering and feeding data as well as instructions among UAV nodes and the base station, the data transmitted is the most important facet of the mission. There is a wide range of methods for securing the payload of network packets transmitted within a network. For example, deep packet inspection and network intrusion detection systems are some of the more frequently encountered security strategies in this regard [31]. These systems operate either by searching for the presence of any predefined strings/signatures [15,32] or by analysing network activities to determine anomalies [33,34]. However, these methods rely on scanning each and every byte of the payload against a large predefined set of strings or signatures that correspond to known attacks. Since the location of such strings within the payload is not deterministic, they require the ability and computational power to search for strings of varying lengths and at random locations within the packet payloads. Furthermore, such systems need to consider the rate at which they can perform these checks in order to avoid network bottlenecks [35].

Considering the aforementioned factors and the resource-constrained as well as the dynamic nature of FANETs and its nodes, such security strategies are deemed unsuitable to be used for payload authentication. Therefore, instead of dealing with string matching or anomaly detection, verifying the payload through the use of data hiding appears to be an attractive alternative. Specifically, in order to avoid creating network bottlenecks, a data hiding venue capable of simultaneously transmitting the condensed payload along with the data frames can be utilized. Typically, network data hiding venues are categorized into two classes, namely, storage and timing channels. In the case of storage channels, the sender embeds selected bits into used fields or other storage locations to transmit (hidden) data [36]. On the other hand, in the case of timing channels, the sender manipulates time delays or packet order to convey secret data [37]. Table 1 records the capacities of some known data hiding venues. While the aforementioned methods are venue-oriented where the authors explore new locations and methods to transmit secret bits (messages), another means of securing communications in a FANET is by completely hiding the transmission itself. Researchers are now looking into ways they can optimize the physical layer parameters of a transmission to ensure that it stays hidden from potential adversaries. Some examples of such works are [38–41], where the authors have explored the fundamental limits of hiding wireless transmissions from a predefined warden. Although the aforementioned methods can hide data in FANET networks, their targeted application is secret communication, hence the utmost important criterion is *secrecy*. On the other hand, our current work focuses on authentication, where we use data hiding at the FANET physical layer to embed an encoding of the payload of each data frame.

Watermarking is another branch of data hiding used for protecting message integrity. Guo et al. [45] proposed a fragile watermarking scheme that utilizes hashing to preserve the integrity of data streams, while Agrawal et al. [46] and Kamel et al. [47] proposed similar watermarking schemes for copyright and data integrity protection of relational databases. Although these methods provide the necessary preservation of data integrity, certain resource constraints have not been considered, for example limited bandwidth, processing power and dynamic mobility. Security mechanisms must be optimized to

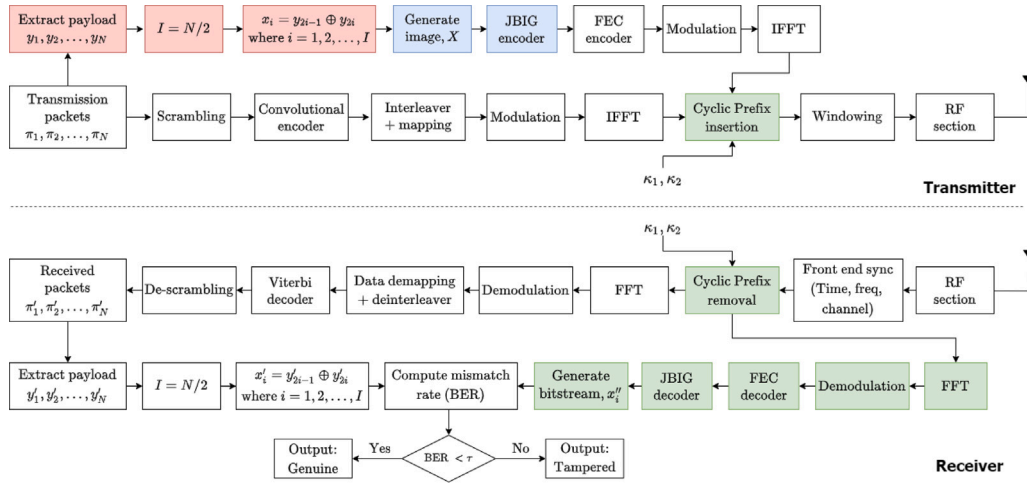


Fig. 1. Flow of processes at the transmitter and receiver for the proposed payload authentication scheme. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

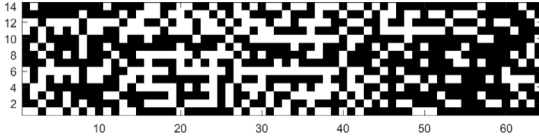


Fig. 2. Sample image generated from bit stream $\{x_i\}_i$. The image is transposed for horizontal display.

Table 1
Hidden data capacity.

Venues	Capacity (bits/frame)
On/Off timing channel [42]	1
Inter-packet gap [43]	96
Rate field [44]	400
Cyclic prefix [29]	3600
OFDM padding [25]	28,800

operate within these constraints, using efficient and lightweight algorithms. On the other hand, our work focuses on the development of a lightweight and efficient scheme that uses a simple data-hiding venue and operations that are not computationally intensive to ensure payload integrity.

Our proposed data hiding scheme for payload authentication is detailed in the next section.

3. Proposed method

The proposed payload authentication method can be broadly divided into four phases, namely: *signature generation*, *signature insertion*, *signature extraction*, and *signatures comparison*. An overview of the proposed scheme is illustrated in Fig. 1. The major steps are outlined below, followed by descriptions of each major step.

3.1. Signature generation

Let $\pi_1, \pi_2, \dots, \pi_N$ be the stream of packets to be transmitted by a legitimate sender. Let $I = N/2$, and divide the packet payload into two groups. Denote the payloads¹ in the packets by y_{2i} and y_i for $i = 1, 2, \dots, I$. Compute the authentication data (i.e., *signature*) $x_i = y_{2i-1} \oplus y_{2i}$, where $\oplus$ denotes the XOR operation. This process is highlighted in red in Fig. 1.

¹ Payloads of each group are extracted at the transport layer.

The resulting bit stream, $\{x_i\}_i$ from step is reshaped to generate a binary image X of dimension $C \times R$ is generated, where $I \leq R \times C$. For example, a payload size of 896 bits can be reshaped to generate an image of 64×14 , as shown in Fig. 2 (transposed to allow a horizontal presentation). Image X will then undergo JBIG2 compression to produce a condensed bit stream of 120 bits. We chose JBIG2 because it can compress any binary image in a lossless manner and its MQ encoder and decoder are efficient [48,49]. These processes are highlighted in blue in Fig. 1.

3.2. Signature insertion

The proposed scheme modifies each Cyclic prefix (CP) between two consecutive OFDM symbols to embed the compressed bit stream that encodes the signature of the packet payload. Specifically, it only uses a quarter of each CP region to embed data, i.e., $P/4$ bits where P is the length of the CP. The basis for not using the entire CP region is because the fading channel taps of the channel are less than or equal to half of a length of a cyclic prefix, i.e., $P/2$ [50]. Therefore, the effect of fading on the embedded data (i.e., error) can be reduced. The modulation mapping of the embedded data follows the same process as that of the actual payload data, using the configurations stipulated by the upper layers of the network stack. Since only $P/4$ bits from the CP region are utilized to embed the authentication data, the starting position (SP) for embedding can be guided by a pseudo-random number generator seeded by a key κ_1 . Similarly, the embedding locations (EL) can be spread across the entire CP region (i.e., non-contiguous chunk), and the order of embedding can also be randomized (ER). The aforementioned processes can be implemented to further enhance the security feature of the proposed payload authentication scheme.

3.3. Signature extraction

At the receiver's end, the CPs in the data frame are extracted after the frame synchronization process. Then, the embedded data segments of the CP are extracted and stored, before being assembled and processed further. In the event the extra security features SP, EL and ER are implemented, the same key κ_1 is required to extract the embedded authentication data. The remainder of the processing at the receiver's end includes channel estimation, de-mapping, demodulation and JBIG2 decoding, to produce bit stream x'_i . The data bits from the output of the demodulation block are transmitted to the transport layer after passing through the de-scrambler. Step G1 will be repeated for the received packets in order to generate bit stream x'_i . It will be compared to the bit stream, x'_i , extracted from CPs to confirm the integrity of the received packet payloads. CP insertion and extraction are highlighted in green in Fig. 1.

3.4. Signatures comparison

At the receiver's end, the authentication data (i.e., signature) x'_i (equivalently, the binary image X') is derived from the received FANET packets. Another copy of the authentication data, denoted by x''_i , is extracted from CPs. Subsequently, the bit error rate (BER) between x'_i and x''_i is computed and compared. When $\text{BER}(x'_i, x''_i) \geq \tau$, we conclude that the packets have been tampered with. Otherwise, we confirm that the received packets are *genuine*. The value for the threshold τ will be empirically decided. In addition, by studying the BER for each CP, we can pinpoint which part of the transmission has been tampered with, which could assist in further analysis and forensics.

4. Enhancements

In this section, two enhancements are proposed to improve the proposed payload authentication system.

4.1. Key-based authentication

One may argue that an adversary can simultaneously modify both the payloads as well as the CPs to pretend that a genuine transmission have occurred (i.e., masquerade attack). To address this potential attack, a key κ_2 can be introduced in the XOR operation in the *Signature Generation* phase (see Section 3.1). The key can be shared prior to the communication, or they can be exchanged using a key exchange scheme such as Diffie–Hellman [51]. Here, the key κ_2 can be considered as a one-time-pad sequence, or it can be utilized to generate a binary sequence to implement a stream cipher. The same approach needs to be performed in the *Signature Extraction* phase (see Section 3.3).

4.2. Symbol time randomization

Since the main aspect of this scheme is to modify the CP, it is important to protect the CP against any possible attacks. If an adversary were to suspect that data has been embedded in the CP, they will be able to successfully overcome the proposed scheme with considerable ease. Therefore, the CP needs to be protected from eavesdropping attacks.

One way of securing the CP is to suppress its periodicity so that it is difficult for the adversary to identify the CPs from the OFDM symbols. In order suppress the cyclic features stemming from periodicity of CPs, a symbol time randomization is put forward. The symbol time is changed by a random index by altering the number of QAM-symbols carried by each symbol. Changing the symbol time can be achieved by varying the number of subcarriers for each symbol. For example, if the symbol time is extended by a positive number, more subcarriers are allocated for the given band because more QAM-symbols are transmitted in parallel during the symbol duration, and vice versa. This leads to the suppression of the periodicity of CP allocation, making it difficult for the adversary to identify which part of the transmission is a CP or an OFDM symbol.

5. Performance evaluation

5.1. FANET simulation model

The simulation of a FANET framework is carried out on Matlab (Ver. R2018b), where the VANET-Simulation-in-MATLAB package downloaded from [52] is modified according to the FANET parameters with reference to [53], which are summarized in Table 2. The 3D Smooth Random Waypoint mobility model was adopted by modifying [53] to generate the UAV movement trajectory, which is different from the dedicated traffic model adopted in [52] to represent an urban setting suitable for a VANET. The simulation boundaries were set to 2000 m $\times$ 2000 m $\times$ 200 m to represent the 3D motion of the

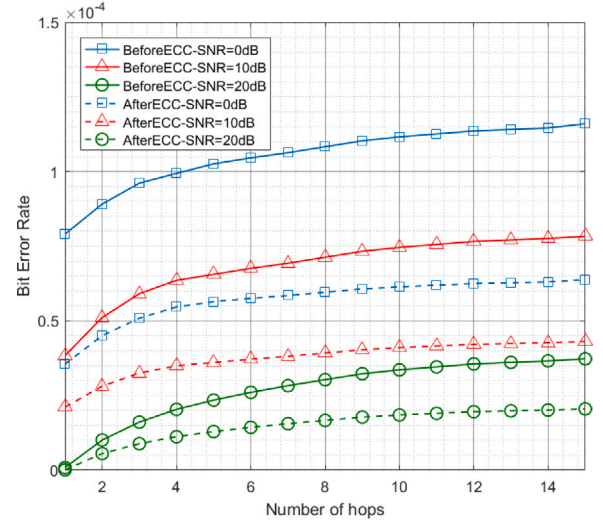


Fig. 3. The graph of BER against number of hops before and after applying error correction coding (Hamming code - 4 data, 3 parity). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

UAV nodes. The number of nodes to be simulated is set to 20, the motion speed of the UAV nodes is set to vary from 20 to 50 m/s, and the simulation time is set to 200 s. A travel time of 10 s is set for a single 3D random walk. For the purpose of our simulations, an urban scenario with a low altitude, crowded environment is considered due to the higher possibility of the existence of potential adversaries and consequently, more node-hijacking events. Therefore, the radio propagation model applied was the Shadowing propagation model with Rayleigh fading [54], viz., the type of traffic is Constant bit rate, and each packet size is set to 512 Bytes. The communication range for each node is set to 500 m.

The chosen routing protocol is Ad Hoc On Demand Vector (AODV) routing, which is a position-based routing protocol. Here, each node is aware of the position information of the all other nodes, and establishes the next forwarding node using the position information. Therefore, each node is required to be equipped with precise GPS devices and each node stores a routing table, which contains a single record for each destination. The entire network is stationary until a connection needs to be established. This routing protocol is therefore, suitable for high speed and dynamic networks [58].

5.2. Bit error rate analysis

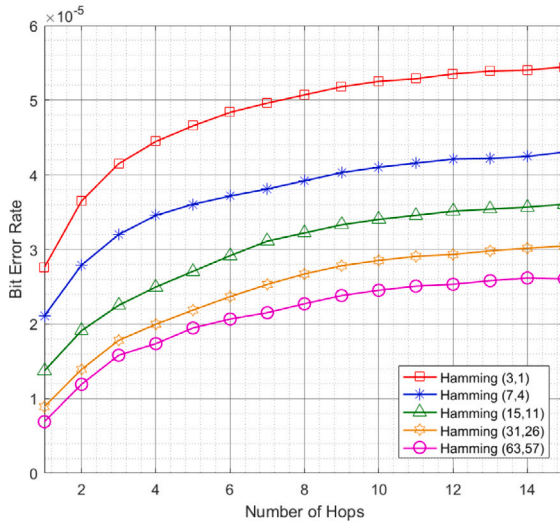
The proposed payload authentication scheme is implemented within the FANET framework detailed in Section 5.1. Experiments are conducted for different hop-counts, ranging from 2 to 15 in a UAV network of 20 nodes. The Bit Error Rate (BER) is calculated from the bit stream, $\{x_i\}_i$ extracted from the modified CP. The results are shown in Fig. 3, where the signal-to-noise ratios (SNR) is set at 0 dB (blue), 10 dB (red) and 20 dB (green). The results show an expected increasing trend in the BER as the hop-count increases, and vice versa. In addition, results also suggest a decreasing trend in the BER as SNR increases, which is also an expected outcome. However, due to the dynamic nature of the FANET and resulting frequent topology changes, bit error is inevitable. Therefore, for the proposed scheme to remain functional, some form of error correction is necessary, and Hamming code [59,60] is applied. Fig. 3 includes the results before and after implementing the Error correction coding (ECC) of 7-bit Hamming code (4 data, 3 parity). It is apparent that the results for all three SNRs show consistent and significant improvement in terms of BER (i.e., lower) after applying H(7,4).

Table 2
Simulation parameters.

Simulation parameters	VANET	FANET
Simulation platform	MATLAB	MATLAB
Simulation area	2000 m × 2000 m	2000 m × 2000 m × 200 m
Mobility model	Dedicated traffic model	3D Random waypoint [55]
MAC layer protocol	IEEE 802.11p	IEEE 802.11p [56]
Type of traffic	Constant bit rate	Constant bit rate
Routing protocol	Dest-sequenced distance vector [57]	Ad Hoc On Demand Vector [58]
Simulation duration	200 s	200 s
Data packet size	32 bytes/packet	512 bytes/packet
Node speed	2–15 m/s	20–50 m/s
Node number	20	20

Table 3
Execution time breakdown for related schemes.

Operations	Time (μs)	Ours	Rajasekaran et al. [18]	Tian et al. [7]	Alladi et al. [19]	Zhang et al. [20]	Hussain et al. [21]
XOR	4.42	2	0	0	9	0	0
Addition	13.53	0	0	1	0	0	0
M/D	95.56	2	0	1	0	0	0
Exponential	53.62	0	7	1	0	0	0
PRNG	1.7	2	0	0	2	0	0
Hash	716.15	0	2	4	0	17	22
HMAC	2400	0	0	0	2	0	0
Pairing	118.48	0	2	0	0	0	0
Fuzzy extraction	776.8	0	0	0	0	0	1
JBIG2	880.72	2	0	0	0	0	0

**Fig. 4.** Graph of BER vs. number of hops for different types of Hamming codes when SNR is set at 10 dB.

To investigate further, we show the performance of different Hamming codes in Fig. 4 when the SNR is set at 10 dB. The results show an improvement in performance in terms of BER when the Hamming codes are scaled up for larger data lengths, and vice versa. However, this creates a large overhead due to interspersing of the redundancy bits (e.g., the actual payload for H(15,11) is 11/15). In addition, the encoding and decoding of Hamming codes also contribute to the storage and computational overhead. Therefore, the user needs to decide and strike a balance while considering the trade-off among payload, complexity, and BER.

Nonetheless, based on the obtained results, we recommend to set the threshold value of $\tau = 0.00012$ and 0.00007 for the non-ECC and ECC settings, respectively, when deciding whether a transmission is genuine or not.

5.3. Execution time

In order to evaluate the efficiency of the proposed payload authentication scheme, the total execution time is determined and compared with other related message authentication schemes proposed for UAV environments. The total execution time is defined in terms of the execution times of significant operations involved in the mentioned schemes. The operations discussed are bit-XOR, one-point addition, multiplication/division, exponential, pseudo-random number generation (PRNG), hashing, hash-based message authentication codes (HMAC), pairing, reverse fuzzy extraction and JBIG2 encoding/decoding for binary images operations. The execution time representations of the aforementioned operations were obtained using an Intel Core i5-8250U Processor 1.6 GHz (6M Cache, up to 3.4 GHz, 4 cores). The execution time of cyclic prefix data hiding is not taken into account because cyclic prefix insertion is a regular part of network operation. The proposed work is compared with related existing schemes such as Rajasekaran et al. [18], Tian et al. [7], Alladi et al. [19], Zhang et al. [20] and Hussain et al. [21] schemes and are included in Table 3. As seen in Fig. 5, it is clear that the proposed payload authentication scheme has a lower total execution time when compared to related work, making it suitable to be operated in the FANET environment.

5.4. Effect of symbol randomization on BER

A careful reader may question whether or not the process of symbol randomization (discussed in Section 4.2) affects the overall BER of the extracted bit stream. Therefore, an experiment is carried out to investigate the effect of symbol randomization on BER. Fig. 6 shows the comparison of the BER before (black) and after (red) symbol randomization for three SNR settings. As observed from the results, the introduction of symbol randomization only leads in a slight increase in BER, which is $\leq 0.006\%$ of the original BER. Since the increase in BER is negligible, we conclude that the inclusion of the symbol randomization process does not impair the performance of the proposed authentication scheme.

5.5. Inference results under different scenarios

In order to investigate the viability of the proposed FANET packet payload authentication scheme, we consider the following scenarios:

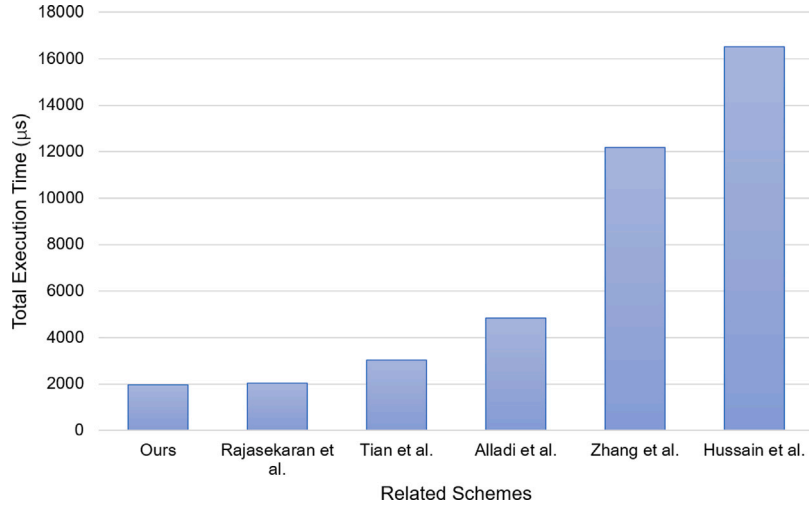


Fig. 5. Total execution time (ms) for different schemes.

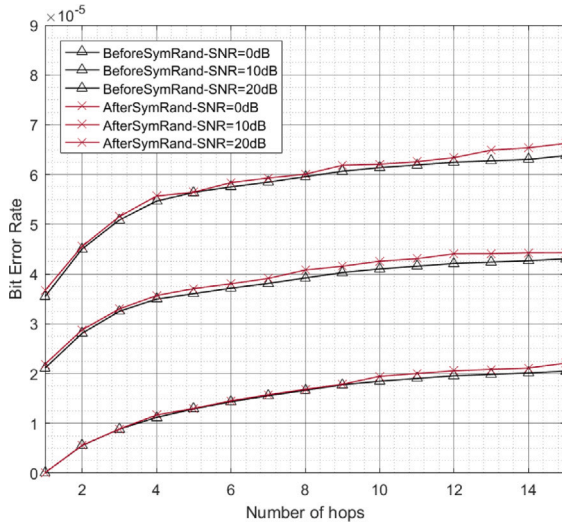


Fig. 6. The graph of BER vs. number of hops before and after applying symbol randomization. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

- Case 1: No tampering has occurred.
- Case 2: The wrong key κ' is used to determine the CP locations to extract the embedded authentication data.
- Case 3: 100% tamper ratio, i.e., all bits of the data frame payload have been tampered with (i.e., flipped).
- Case 4: 50% tamper ratio.
- Case 5: 25% tamper ratio.
- Case 6: 10% tamper ratio.
- Case 7: Three selected data frames have been tampered with.

For experiment purposes, we set the signal-to-noise ratio at 10 dB. Fig. 7 shows the BER results for cases 1–6. The threshold value was set to $\tau = 0.00007$ for ECC setting (as discussed in Section 5.2) in order to determine if the data frames transmitted are genuine. As seen from the results, Cases 2–6 yield significantly higher BER values, which is well above the threshold τ . These are expected outcomes because tampering has occurred. Hence, for the scenarios attributed to Cases 2–6, the data frames would be rejected by the proposed payload authentication scheme.

To demonstrate the localization capability of the proposed payload authentication scheme, Fig. 8 shows the BER results for Case 7, where

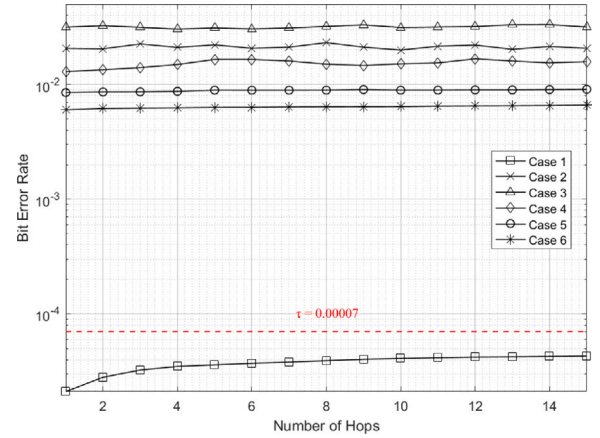


Fig. 7. The graph of BER vs. number of hops for cases 1–6.

three selected data frames are tampered with. Specifically, the 15th, 45th and 85th data frames are selected, where the payloads of these data frames are replaced with random bit streams of the same length. As seen in Fig. 8, the BER for data frame number 15, 45 and 85 spike and the values are significantly higher (by an order of 100) in comparison to the untampered data frames. These spikes confirm that the tampered network packets can be identified (i.e., localized), at the granularity level of specific data frames in a single transmission, which would be useful in forensic investigation.

6. Conclusion

FANETs, while being abundantly popular in recent years due to its ease of operation, versatility and low cost, are also extremely vulnerable to a number of security threats. This paper presents a payload verification scheme that essentially condenses the payload of a physical layer data frame into a compact bit sequence (i.e., signature) and embeds the signature into the cyclic prefix at the physical layer before transmitting through the network. To ensure that the scheme is lightweight and energy efficient, a simple XOR operation paired with image generation and JBIG2 compression are used to generate the bit stream to be embedded into the CP. At the receiver, the hidden data is extracted and decoded to be compared against the payload of the received data frame. The proposed scheme was evaluated in terms of bit error rate and was found to produce BERs of less than 0.7×10^{-4}

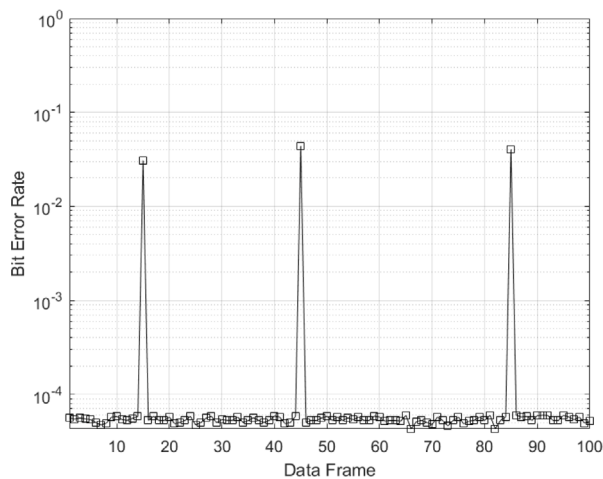


Fig. 8. The graph of BER vs. data frame number. The 15th, 45th, and 85th data frames are tampered with using randomized bit stream to illustrate the localization capability of the proposed payload authentication method.

with the help of the implementation of 7-bit Hamming code. Several experiments are conducted under different scenarios in order to test the viability and localization capability of the proposed scheme.

The results show that it is able to successfully determine whether a given data frame is genuine or not. In addition, it is also capable in localizing the tampered data which is useful in forensic investigation. Furthermore, two enhancements have also been put forward to further protect the proposed payload authentication scheme from eavesdropping and man-in-the-middle attacks. The proposed scheme has an inherent capacity limitation. Specifically, each data frame can support a maximum of 3600 bits of CP locations. Since only a fraction of the total locations (i.e., $P/4$) are embeddable, the compressed bitstream that can be accommodated by the CP is at most 900 bits. Consequently, with a compression ratio of 14.93:1 (achieved for binary images during evaluation), the size of the data frame payload must be restricted to approximately 3360 bytes to ensure data integrity. As per the IEEE 802.11p standards, the typical data frame size varies from 64 to 1500 bytes [61]. Therefore, the capacity of the proposed authentication scheme remains sufficient for the considered scenario.

As future work, the proposed scheme can be paired with a user-authentication scheme in order to mutually authenticate the sender and receiver to add an additional layer of security. The threshold ($\tau = 0.00007$) has been defined to account for the bit error rate that is evident during FANET communication. It would not be possible to detect a modification of only a few bits due to this reason. As future work, the proposed scheme can be further investigated with no quantization and a threshold value of 0 to maximize the level of security (i.e., zero tolerance for error), along with an added extension to deal with the relatively high amount of bit errors naturally occurring in this environment. In addition, we will execute the proposed scheme on actual UAV processors to evaluate its viability. This implementation will serve as a crucial testbed to determine the efficacy and efficiency of the proposed scheme in the UAV network environment under realistic operational conditions, including various types of UAVs, different network topologies, and varying traffic loads. The results of this investigation will provide insights into the potential practicality of the proposed scheme in real-world UAV networks and aid in its refinement and optimization.

CRedit authorship contribution statement

Dilshani Mallikarachchi: Data Hiding/security, Communication, Experiment and data collection, Analysis and discussion, Manuscript

writing. **KokSheik Wong:** Data Hiding/security, Analysis and discussion, Manuscript writing. **Joanne Mun-Yee Lim:** Communication, Analysis and discussion, Manuscript writing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article

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